HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **eBPF**-based packet filtering and routing logic for a high-throughput edge traffic platform, processing approximately 50,000 requests per second.
- Managed the migration of a critical legacy traffic management system to a modern **Envoy** and **Rust**-based architecture, ensuring zero production downtime during the transition.
- Automated the generation of **production runbooks** and troubleshooting guides using Python scripting, reducing incident resolution time by an average of 22% for common issues.
- Developed and maintained **Go** microservices responsible for API gateway functionality and request authentication, handling over 100 million user requests daily.
- Addressed numerous **incident-action-item** backlogs by identifying and resolving root causes in distributed systems, improving overall platform stability.
- Collaborated with security teams to integrate a **WAF** implementation (Coraza) for real-time threat detection and mitigation at the network edge.
- Containerized platform services using **Docker** and orchestrated deployments onto a Kubernetes cluster (EKS), managing resource allocation and scaling policies.

SKILLS

Edge Traffic & Security – Envoy, eBPF, HAProxy, WAF, Coraza, ModSecurity, DDoS Mitigation, Rate Limiting, TLS/mTLS Termination

Programming Languages – Rust, Go, C++, C, Python

Distributed Systems & Networking – TCP/IP, HTTP, DNS, L4, L7 Networking, CDNs, Load Balancers, Routing Matrices

Cloud & Orchestration – AWS, EKS, Docker, Kubernetes

Operations & Reliability – Production Runbooks, Incident Response, Root Cause Analysis, CI/CD, Jenkins, Git

PROJECTS

Edge Traffic Platform Modernization (HCL Technologies)

- Engineered a new edge traffic platform using **Go** and **Envoy**, replacing legacy components and improving global traffic flow for enterprise customers.
- Implemented dynamic routing rules and load balancing strategies, optimizing request distribution and achieving a 15% reduction in latency under peak loads.
- Developed automated deployment pipelines with **Jenkins** and **Docker**, facilitating consistent and reliable releases of platform updates.

eBPF-based Network Security Module (MSc Project)

- Developed an **eBPF** program in **Rust** to monitor and filter network traffic at the kernel level, enhancing security for distributed systems.
- Implemented custom logic for anomaly detection and threat identification, successfully identifying simulated DDoS attacks with 92% accuracy in test environments.
- Integrated the eBPF module with a user-space control plane built with **Python** for configuration management and reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)